



**Website:**

[www.questions.maarula.in](http://www.questions.maarula.in)

हे शारदे माँ

**Mathem solvex  
for PYQ**

**MAARULA CLASSES**

# CUET PG 2022

## OFFICIAL PAPER

 **Topic Wise Weightage &  
Detailed Trend Analysis**

 **All Last Years Papers**

 **Syllabus & Important  
Documents**

 **Topic Wise All PYQ's  
CUET-PG – FREE TEST**

 **Und  
Ami**

[https://youtube.com/@maarulaclasses?  
si=XkKXyJ\\_2a16NiK-K](https://youtube.com/@maarulaclasses?si=XkKXyJ_2a16NiK-K)



TAB THIS



[https://youtube.com/@maarulaclasses?  
asses?si=XkKXyJ\\_2a16NiK-K](https://youtube.com/@maarulaclasses?si=XkKXyJ_2a16NiK-K)

**AMIT KATIYAR**  
(MCA-JNU)



0512-3163515



9935985550



01. From among the four options given, choose the correct sequence of the four phrases given below, to make a meaningful sentence:

A. Lower our guard and our masks  
B. Repeatedly weather the pandemic might finally  
C. Be over, many of use are willing to  
D. Having asked ourselves

(a) A, C, B, D (b) D, B, C, A  
(c) C, D, A, B (d) B, A, D, C

### ENGLISH – Sentence Arrangement

02. Which of the following is one-word substitute for 'the quality of having a ready insight into things'?

(a) Felicity (b) Tenacity  
(c) Perspicacity (d) Chromaticity

### ENGLISH – One-Word Substitution

03. Identify the correct indirect narration for the following sentence:

**The lady said to the servant, "If you don't wash the clothes properly, I will dismiss you".**

(a) The lady warned the servant that she would dismiss her if she didn't wash the clothes properly.  
(b) The lady told the servant that she would dismiss her on the event of bad work.  
(c) The lady cautioned the servant that she must wash the clothes properly  
(d) The lady advised the servant to wash the clothes properly

### ENGLISH – Indirect Speech

04. Pick the plural from of 'matrix'

(a) Matrixes (b) Matrices (c) Matries (d) Matrii

### ENGLISH – Plural Form

05. Which of the following options will be correct choice for the blank?

I'm not sure where he is. He \_\_\_\_\_ out for a walk.

(a) gone (b) may have gone  
(c) has gone (d) is going

### ENGLISH – Tense

06. Match List I with List II:

|    | List I    |      | List II |
|----|-----------|------|---------|
|    | (Animals) |      | (Sound) |
| A. | Apes      | I.   | Howl    |
| B. | Wolves    | II.  | Gibber  |
| C. | Lambs     | III. | Squel   |
| D. | pigs      | IV.  | Bleat   |

Choose the correct answer from the options given below:

(a) A – II; B – I; C – IV; D – III  
(b) A – III; B – IV; C – II; D – I  
(c) A – IV; B – II; C – III; D – I  
(d) A – II; B – III; C – IV; D – I

### ENGLISH – Match (Animals–Sound)

07. From among the four options given, choose the one which is a grammatically correct sentence.

(a) Frequent entertain or equality time in home calling out for comfort seating.

- (b) Frequent entertaining or quality time at home calls out for comfortable seating.  
(c) Comfort seat calling out for frequent entertaining or quality time at home.  
(d) Frequent entertaining calls out for comfortable seating or quality time at home.

### ENGLISH – Sentence Correction

08. From among the four options given, choose the one in which the word has been spelt correctly.

(a) Vituperative (b) Vetuperative  
(c) Vituparative (d) Vituperetive

### ENGLISH – Spelling

09. Choose the correct option to make a meaning full sentence.

Shena is \_\_\_\_\_ in the art of cooking.

(a) Adept (b) Adopt  
(c) Adapt (d) Admit

### ENGLISH – Vocabulary

10. Choose the correct preposition to complete the following sentence:

**Ramanujan is good \_\_\_\_\_ mathematics.**

(a) at (b) in  
(c) for (d) into

### ENGLISH – Preposition

11. Reversing Roe is a documentary film centered on:

(a) Civil Rights Movement  
(b) Debate on Abortion Rights  
(c) Women's Right to Study Medicine  
(d) Voting Rights for the Blacks

### GK – Documentary / Current Affairs

12. Match List I with List II:

| List I                |                         | List II        |           |
|-----------------------|-------------------------|----------------|-----------|
| (Noble Prize Winners) |                         | (Area/Subject) |           |
| A.                    | Md. Yunus               | I.             | Chemistry |
| B.                    | Abdus Salam             | II.            | Peace     |
| C.                    | Venkatraman Ramakrishna | III.           | Economics |
| D.                    | Marie Ressa             | IV.            | Physics   |

Choose the correct answer from the options given below:

(a) A – I; B – II; C – III; D – IV  
(b) A – III; B – IV; C – I; D – II  
(c) A – II; B – III; C – IV; D – I  
(d) A – IV; B – I; C – II; D – III

### GK – Nobel Prize

13. Match List I with List II:

| List I     |              | List II    |                              |
|------------|--------------|------------|------------------------------|
| (FM Radio) |              | (Tag line) |                              |
| A.         | Radio Mirchi | I.         | Dhoon badal ke toh Dekho     |
| B.         | My FM        | II.        | International Indians        |
| C.         | Big FM       | III.       | Jio Dil Se                   |
| D.         | Ratio One    | IV.        | ..... Sunnewale always Khush |



Choose the correct answer from the options given below:

- (a) A – I; B – II; C – III; D – IV  
(b) A – III; B – IV; C – II; D – I  
(c) A – IV; B – III; C – I; D – II  
(d) A – II; B – I; C – IV; D – III

## GK – FM Radio Tagline

14. Match List I with List II:

|    | List I         |      | List II                 |
|----|----------------|------|-------------------------|
|    | (Author)       |      | (Title of the book)     |
| A. | Salman Rushdie | I.   | The Inheritance of Loss |
| B. | Arundhati Roy  | II.  | The White Tiger         |
| C. | Kiran Desai    | III. | The God of Small Things |
| D. | Arvind Adiga   | IV.  | Midnight's Children     |

Choose the correct answer from the options given below:

- (a) A – I; B – II; C – IV; D – III  
(b) A – III; B – I; C – II; D – IV  
(c) A – II; B – IV; C – III; D – I  
(d) A – IV; B – III; C – I; D – II

## ENGLISH – Books & Authors

15. Which company manufactured the Ambassador car?

- (a) Tata Motors Ltd. (b) Mahindra and Mahindra  
(c) Ashok Leyland Ltd. (d) Hindustan Motors Ltd.

## GK – Company / Industry

16. Pointing towards a boy, a girl said, "He is the son of the daughter of the father of my uncle." How is the boy related to the girl?

- (a) Brother (b) Uncle  
(c) Son-in-law (d) Nephew

## REASONING – Blood Relation

17. If A + B means A is the mother of B;  
A – B means A is the brother of B;  
A % B means A is the father of B and  
A × B means A is the sister of B,  
which of the following shows that P is the maternal uncle of Q?

- (a) Q – N + M × P (b) P + S × N – Q  
(c) P – M + N × Q (d) Q – S % P

## REASONING – Coded Relation

18. If Z = 52 and ACT = 48, then RAT will be equal to

- (a) 72 (b) 78  
(c) 92 (d) 64

## REASONING – Alphabet Coding

19. There are equation that have become wrong due to incorrect order of signs. From the four alternatives, given below, find out the correct order of signs. So that equation becomes right.

$$8 \div 9 = 9 - 81$$

- (a) -, -, + (b) +, -, ÷  
(c) =, +, - (d) ×, +, =

## REASONING – Operator Arrangement

20. If + means ×, / means -, \* means /, and - means +, what will be the value of

$$4 + 11 / 5 - 55 = ?$$

- (a) -48.5 (b) -11  
(c) 79 (d) None of these

## REASONING – Coded Arithmetic

21. If  $(25)^{7.5} \times (5)^{2.5} \div (125)^{1.5} = 5^x$  then  $x = ?$

- (a) 13 (b) 8.5 (c) 16 (d) 17.5

## MATHS – Linear Equation

22. Half of a large Pizza is cut into 4 equal-sized pieces, and the other half is cut into 6 equal-sized pieces. If a person were to eat 1 of the large pieces and 2 of the smaller pieces, what fraction of the pizza would remain uneaten?

- (a)  $\frac{5}{12}$  (b)  $\frac{1}{24}$  (c)  $\frac{17}{24}$  (d)  $\frac{13}{24}$

## MATHS – Fractions

23. In a business, A and C invested amounts in the ratio 2:1, whereas the ratio between amounts invested by A and B was 3:2. If Rs. 1,57,300 was their profit, how much amount did B receive?

- (a) Rs. 24,200 (b) Rs. 72,600  
(c) Rs. 36,300 (d) Rs. 48,400

## MATHS – Partnership

24. Company A produces toy trucks at a cost of Rs. 50.00 each for the first 100 trucks and Rs. 35.00 for each additional truck. If 500 toy trucks were produced by Company A and sold for Rs. 100.00 each, what was Company A's gross profit?

- (a) Rs. 31,000 (b) Rs. 15,000  
(c) Rs. 21,000 (d) Rs. 13,000

## MATHS – Profit & Loss

25. A shopkeeper fixes the marked price of an item 35% above its cost price. The percentage of discount allowed to gain 5% is.

- (a) 43% (b) 27%  
(c) 22% (d) 31%

## MATHS – Discount

26. Choose the odd one out:

- (a) Mohammed Rafi (b) K. L. Saigal  
(c) Naushad (d) Adnan Sami

## GK – Odd One Out (Music)

27. Which of the following options is the correct alphabetical order of the four words given as:

## Rumbustious, Rumanian, Rumour, Rumble

- (a) Rumanian, Rumble, Rumour, Rumbustious  
(b) Rumour, Rumbustious, Rumble, Rumanian  
(c) Rumanian, Rumbustious, Rumour, Rumble  
(d) Rumble, Rumour, Rumbustious, Rumanian

## REASONING – Alphabetical Order

28. If 'ORANGE' is coded as 'MTYPEG' in a certain language, then how will 'CHILLY' be coded in that language?

- (a) AJGNNW (b) AJGNJA  
(c) AJKNNW (d) AJGOOW

## REASONING – Coding-Decoding

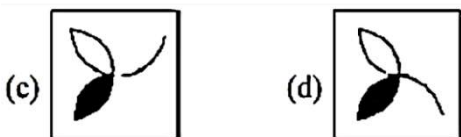
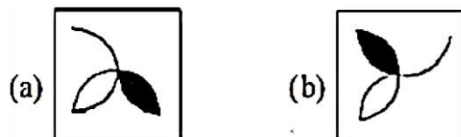
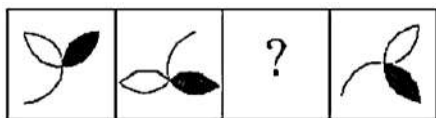
29. Choose the odd one out from the following

- (a) Morarji Desai (b) V. P. Singh  
(c) Narendra Modi (d) Sardar Patel

## GK – Odd One Out (Politics)



30. Which will be the missing patten in the series? Redraw figure



### REASONING – Figure Series

31. If 'MANGO' is coded as 'KCLIM' in a certain language then how will 'SWEETY' be coded in that language?

- (a) QYCGRA (b) QYCCRA  
(c) QYGCRA (d) YQCGRA

### REASONING – Coding-Decoding

32. Which among the following words comes first in a dictionary?

**Shrub, Shudder, Shroud, Shuttle**

- (a) Shrub (b) Shuttle  
(c) Shroud (d) Shudder

### REASONING – Dictionary Order

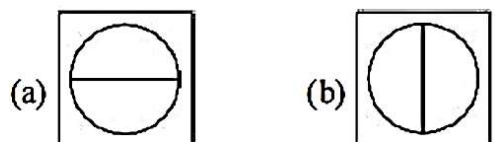
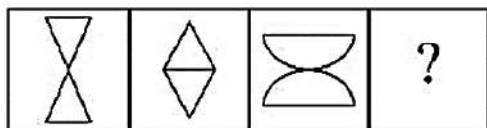
33. Which of the following options is the correct alphabetical order of the four words given below?

**Sport, Squash, Sporadic, Sprout**

- (a) Sport, Sprout, Sporadic, Squash  
(b) Sporadic, Sport, Sprout, Squash  
(c) Sporadic, Sport, Squash, Sprout  
(d) Squash, Sprout, Sport, Sporadic

### REASONING – Alphabetical Order

34. What will be the Next Pattern in the series?



### REASONING – Figure Pattern

35. Complete the series by filling the missing number

**2, 5, 14, 57, 284, ?**

- (a) 1420 (b) 1421 (c) 1704 (d) 1705

### MATHS – Number Series

36. If  $A = \begin{bmatrix} \cos B & -\sin B \\ \sin B & \cos B \end{bmatrix}$  then  $A + A^T = I$  for B equals to

- (a)  $\frac{\pi}{3}$  (b)  $\frac{\pi}{6}$  (c)  $\pi$  (d)  $\frac{3\pi}{2}$

### MATHS – Trigonometric Equation

37. The number of 7-digit numbers whose sum of the digits equals to 10 and which is formed by using the digit 1, 2 and 3 only is

- (a) 55 (b) 66  
(c) 77 (d) None of these

### MATHS – Permutation & Combination

38. If from each of the three boxes containing 3 white and 1 black, 2 white and 2 black, 1 white and 3 black balls, one ball is drawn at random, then the probability that 2 white and 1 black balls will be drawn is

- (a)  $\frac{13}{32}$  (b)  $\frac{1}{4}$  (c)  $\frac{1}{32}$  (d)  $\frac{3}{16}$

### MATHS – Probability

39. Let  $f(x) = |x| - 1$  then point (s) where  $f(x)$  is not differentiable is (are):

- (a) 0,  $\pm 1$  (b)  $\pm 1$  (c) 0 (d) 1

### MATHS – Differentiability

40. Let  $f : [2, \infty] \rightarrow R$  be the function defined by  $f(x) = x^2 - 4x + 5$ , then the range of  $f$

- (a) R (b)  $[1, \infty)$   
(c)  $[4, \infty)$  (d)  $(-\infty, 0]$

### MATHS – Functions (Range)

41. The function  $f(x) = \frac{[\ln(1+ax)] - \ln(1-bx)]}{x}$  is not defined at  $x = 0$ . What value may be assigned to  $f$  at  $x = 0$ , so that it is continuous?

- (a)  $a + b$  (b)  $a - b$   
(c)  $b - a$  (d)  $\ln a + \ln b$

### MATHS – Continuity

42. The area enclosed between the graphs of  $y = x^3$  and the lines  $x = 0$ ,  $y = 1$ ,  $y = 8$  is

- (a) 7 (b) 12  
(c)  $\frac{45}{4}$  (d)  $\frac{21}{8}$

### MATHS – Area Under Curve

43. If the vertices of a triangles are  $O(0, 0)$ ,  $A(a, 0)$  and  $B(0, a)$ . Then, the distance between its circumcenter and orthocenter is:

- (a)  $\frac{a}{2}$  (b)  $\frac{a}{\sqrt{2}}$   
(c)  $\sqrt{2}a$  (d)  $\frac{a}{4}$

### MATHS – Coordinate Geometry

44. The straight line  $x + y = 0$ ,  $3x + y - 4 = 0$  and  $x + 3y - 4 = 0$  form a triangle which is

- (a) Right angled (b) Equilateral  
(c) Isoscales (d) Isoscales and right angled

### MATHS – Straight Line

45. If one of the lines of  $ax^2 + 2hxy + by^2 = 0$  bisects the angle between the axes in the first quadrant, then

- (a)  $h^2 - ab = 0$  (b)  $h^2 + ab = 0$   
(c)  $(a + b)^2 = h^2$  (d)  $(a + b)^2 = 4h^2$

### MATHS – Pair of Lines



46. What is the value of:  
 $[\tan^2(90 - \theta) - \sin^2(90 - \theta)] \operatorname{cosec}^2(90 - \theta) \cot^2(90 - \theta)$   
 (a) 0 (b) 1 (c) -1 (d) 2

**MATHS – Trigonometric Identity**

47. If  $A + B = 45^\circ$ , then  $(1 + \tan A)(1 + \tan B)$  is equal to  
 (a) 4 (b) 2 (c) 3 (d) 1

**MATHS – Trigonometric Value**

48. If  $\vec{a}$  and  $\vec{b}$  are two unit vectors such that  $\vec{a} + 2\vec{b}$  and  $5\vec{a} - 4\vec{b}$  are perpendicular to each other, then the angle between  $\vec{a}$  and  $\vec{b}$  is:  
 (a)  $45^\circ$  (b)  $60^\circ$   
 (c)  $\cos^{-1}\left(\frac{1}{3}\right)$  (d)  $\cos^{-1}\left(\frac{2}{7}\right)$

**MATHS – Vector Algebra**

49. Let  $a = \hat{i} - \hat{j}$  and  $b = \hat{i} + \hat{j} + \hat{k}$  and  $c$  be a vector such that  $(a \times c) + b = 0$  and  $a \cdot c = 4$ , then  $|c|^2$  equal to:  
 (a) 8 (b)  $\frac{19}{2}$  (c) 9 (d)  $\frac{17}{2}$

**MATHS – Vector Triple Product**

50. If  $\vec{a}, \vec{b}, \vec{c}$  and  $\vec{d}$  are the unit vectors such that  $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 1$  and  $(\vec{a} \cdot \vec{c}) = \frac{1}{2}$ , then  
 (a) only  $\vec{a}, \vec{b}, \vec{c}$  are non-coplanar  
 (b) only  $\vec{a}, \vec{b}, \vec{d}$  are non-coplanar  
 (c) Both  $\vec{a}, \vec{b}, \vec{c}$  and  $a, b, d$  are  $\vec{a}, \vec{b}, \vec{d}$  non-coplanar  
 (d) Both  $\vec{a}, \vec{b}, \vec{c}$  and  $\vec{a}, \vec{b}, \vec{d}$  are coplanar

**MATHS – Coplanarity of Vectors**

51. Let  $A = \{1, 2, 3\}$  and consider the relation  $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (1, 3)\}$  then  $R$  is  
 (a) Reflexive but not symmetric  
 (b) Reflexive but no transitive  
 (c) Symmetric and transitive  
 (d) Equivalence relation

**MATHS – Relations**

52. A spring is being moved up and down. An object is attached to the end of the spring that undergoes a vertical displacement. The displacement is given by the equation  $y = 3.50 \sin t + 1.20 \sin 2t$ . Find the first two values of  $t$  (in seconds) for which  $y = 0$ .  
 (a)  $t = 0, \frac{\pi}{4}$  (b)  $t = 0, \frac{\pi}{2}$   
 (c)  $t = 0, \pi$  (d)  $t = 0, \frac{\pi}{6}$

**MATHS – SHM**

53. A ball is thrown off the edge of a building at an angle of  $60^\circ$  and with the initial velocity of 5 meters per second. The equation that represents the horizontal distance of the ball  $x$  is  $x = v_0 (\cos \theta) t$  where  $v_0$  is the initial velocity,  $\theta$  is the angle at which it is thrown and  $t$  is the time in seconds. About how far will the ball travel in 10 seconds?  
 (a)  $25\sqrt{3}$  m (b)  $50\sqrt{2}$  m  
 (c) 25 m (d)  $\frac{25}{\sqrt{3}}$  m

**MATHS – Projectile Motion**

54. Let  $b$  be a positive integer and  $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a - b = nm \text{ for some } m \neq 0 \in \mathbb{Z}\}$   
 (a) Reflexive on  $\mathbb{Z}$  (b) Symmetric  
 (c) Transitive (d) Equivalence relation of  $\mathbb{Z}$

**MATHS – Relations (Properties)**

55. The  $a, b, c$  and  $d$  are in GP and are in ascending order such that  $a + d = 112$  and  $b + c = 48$ . If the GP is continued with  $a$  as the first term, then the sum of the first six terms is  
 (a) 1156 (b) 1256  
 (c) 1356 (d) 1456

**MATHS – Geometric Progression**

56. Given below are two statements:  
**Statement I:** If  $A \subset B$ ; then  $B$  can be expressed as  $B = A \cup (\bar{A} \cap B)$  and  $P(A) > P(B)$ .  
**Statement II:** If  $A$  and  $B$  are independent events, then  $(\bar{A} \text{ and } \bar{B})$ ,  $(\bar{A} \text{ and } B)$  and  $(A \text{ and } \bar{B})$  are also independent.  
 In the light of the above statements, choose the most appropriate answer from the options given below:  
 (a) Both Statement I and Statement II are true  
 (b) Both Statement I and Statement II are false  
 (c) Statement I is true but Statement II is false  
 (d) Statement I is false but Statement II are true

**MATHS – Probability (Statements)**

57. Given the following truth table:

| A | B | x |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

Which of the following Boolean functions does it represent?

- (a) OR (b) XOR (c) NOR (d) XNOR

**COMPUTER – Boolean Logic**

58. If  $\vec{a}, \vec{b}$  and  $\vec{c}$  are unit vectors, then  $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$  does not exceed:  
 (a) 4 (b) 9 (c) 8 (d) 6

**MATHS – Vector Algebra**

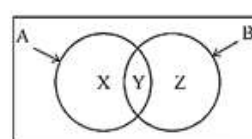
59. The product of two four bit positive binary numbers 1011 and 0011 is \_\_\_\_\_  
 (a) 1000011 (b) 100001  
 (c) 100101 (d) 100111

**COMPUTER – Binary Multiplication**

60. If  $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ ,  $\vec{a} \cdot \vec{b} = 1$  and  $\vec{a} \times \vec{b} = \hat{j} - \hat{k}$  then  $\vec{b}$  is equal to:  
 (a)  $\hat{i} - \hat{j} + \hat{k}$  (b)  $2\hat{j} - \hat{k}$   
 (c)  $\hat{i}$  (d)  $2\hat{i}$

**MATHS – Vector Algebra**

61. Consider the diagram given below and the following two statements:



**Statement I:** Events  $A$  and  $B$  can be expressed as:

$$A = (A \cap \bar{B}) \cup Y \quad B = (A \cap B) \cup Z$$

**Statement II:** Events  $A$  and  $B$  can be expressed as:

$$A = X - Y \quad B = Y + Z$$



In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both Statement I and Statement II are true
- (b) Both Statement I and Statement II are false
- (c) Statement I is true but Statement II is false
- (b) Statement I is false Statement II are true

## MATHS – Set Theory

62. The simplified form of Boolean expression  $AB + AB'$  is \_\_\_\_\_.

- (a) A
- (b) B
- (c)  $1 + A$
- (d)  $1 + B'$

## COMPUTER – Boolean Algebra

63. Representation of  $-11$  in sign – and magnitude is

- (a) 11011
- (b) 010111
- (c) 10011
- (d) 01110

## COMPUTER – Number Representation

64. 1001 is 2's complement representation of \_\_\_\_\_.

- (a)  $-7$
- (b)  $+9$
- (c)  $+6$
- (d)  $-6$

## COMPUTER – 2's Complement

65. Match List I with List II

|    | List I     |      | List II       |
|----|------------|------|---------------|
| A. | $(x + y)'$ | I.   | I             |
| B. | $x + 1$    | II.  | $x' + y'$     |
| C. | $(xy)'$    | III. | $x' \cdot y'$ |
| D. | $x + 0$    | IV.  | $x$           |

Choose the correct answer from the options given below:

- (a) A – II; B – I; C – III; D – IV
- (b) A – II; B – IV; C – I; D – III
- (c) A – III; B – I; C – IV; D – II
- (d) A – III; B – I; C – II; D – IV

## COMPUTER – Boolean Identities

66. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

**Assertion A:** Carbon Monoxide when inhaled causes death.

**Reason R:** Carbon Monoxide combines with haemoglobin.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true and R is not the correct explanation of A
- (c) A is true but R is not false.
- (d) A is false but R is true.

## SCIENCE – Biology (Carbon Monoxide)

67. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

**Assertion A:** In a class of 40 students, 22 drink sprite, 10 drink Sprite but not Pepsi. Then the number of student who drink both Sprite and Pepsi is 15.

**Reason R:** For any two finite sets A and B,

$$n(A) = n(A - B) + n(A \cap B)$$

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true and R is not the correct explanation of A
- (c) A is true but R is not false.
- (d) A is false but R is true.

## MATHS – Set Theory

68. Universities in India are preferring traditional attire during convocations.

- (A) Indian Universities are against western dresses.
- (B) Indian Universities wants to promote traditional dresses.
- (C) Indians love to celebrate occasions traditionally.
- (D) Indian traditional dresses are better than western wear.

(E) Western wear is banned by government.

Choose the correct answer from the options given below.

- (a) B and C only
- (b) A, B and C only
- (c) A, B and D only
- (d) D and E only

## REASONING – Statement Conclusion

69. Match List I with List II

|    | List I                                                                                                               |      | List II        |
|----|----------------------------------------------------------------------------------------------------------------------|------|----------------|
| A. | If 4 <sup>th</sup> term of a G.P. is square of its second term, and its first term is 3, then common ratio is _____. | I.   | 5              |
| B. | The first term of an AP is 5, the last term is 45 and the sum of the terms is 400. The number of term is _____.      | II.  | $\frac{5}{-2}$ |
| C. | The sum of three numbers which are in AP is 27 and sum of their squares is 293. Then the common difference is _____. | III. | 16             |
| D. | The fourth and 54th terms of an AP are, respectively 64 and $-61$ . The common difference is _____.                  | IV.  | 3              |

Choose the correct answer from the options given below:

- (a) A – IV; B – III; C – I; D – II
- (b) A – III; B – II; C – I; D – IV
- (c) A – II; B – III; C – I; D – IV
- (d) A – II; B – I; C – III; D – IV

## MATHS – AP & GP

70. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

**Assertion A:** The system of equations

$x + y + z = 4$ ,  $2x - y + 2z = 5$ ,  $x - 2y - z = -3$  has unique solution.



**Reason R:** If A is  $3 \times 3$  matrix and B is a  $3 \times 1$  non-zero column matrix, then the equation  $AX = B$  has unique solution if A is non-singular.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both A and R are correct and R is the correct explanation of A
- (b) Both A and R are correct and R is not the correct explanation of A
- (c) A is correct but R is not correct.
- (d) A is correct but R is correct.

## MATHS – System of Linear Equations

71. Given below are two statements:

**Statement I:** 011010 is 2's complement representation of -37.

**Statement II:**  $111_{(8)}$  is octal representation of 73.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both Statement I and Statement II are true
- (b) Both Statement I and Statement II are false
- (c) Statement I is true but Statement II is false
- (d) Statement I is false Statement II are true

## COMPUTER – Number System

72. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

**Assertion A:** Binary information is represented in digital in terms of 'zeros' and 'ones'.

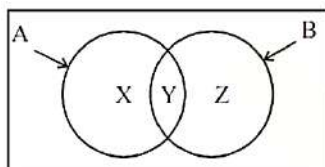
**Reason R:** Digital computers are electronic devices, which operate using electrical voltages.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both A and R are correct and R is the correct explanation of A
- (b) Both A and R are correct and R is not the correct explanation of A
- (c) A is correct but R is not correct.
- (d) A is correct but R is correct.

## COMPUTER – Computer Fundamentals

73. Consider the diagram given below and the following two statements:



**Statement I:** Regions X, Y and Z can be expressed as  $A \cap \bar{B}$ ,  $A \cap B$  and  $\bar{A} \cap B$  respectively..

**Statement II:**  $P(Y) = P(A) - P(X) = P(B) - P(Z)$ .

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both Statement I and Statement II are true
- (b) Both Statement I and Statement II are false
- (c) Statement I is true but Statement II is false
- (d) Statement I is false Statement II are true

## MATHS – Probability

74. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

**Assertion A:** Goitre is a common disease in mountainous regions.

**Reason R:** The diet of the people in mountains lacks Iodine content.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both A and R are correct and R is the correct explanation of A
- (b) Both A and R are correct and R is not the correct explanation of A
- (c) A is correct but R is not correct.
- (d) A is correct but R is correct.

## SCIENCE – Biology (Iodine Deficiency)

75. Which of the following storage hardware can be used as a back-up device?

- (A) HDD
- (B) ROM
- (C) RAM
- (D) Cache
- (E) Magnetic Tape

Choose the correct answer from the options given below:

- (a) only A, B and C
- (b) only B, C and D
- (c) only A and E
- (d) only A, C and E

## COMPUTER – Storage Devices

76. In a class there are 400 students, the following table shows the number of students studying one or more of the subjects:

| Subject                              | Number of students |
|--------------------------------------|--------------------|
| MATHSemantics                        | 250                |
| Physics                              | 150                |
| Chemistry                            | 100                |
| MATHSemantics and Physics            | 100                |
| MATHSemantics and Chemistry          | 60                 |
| Physics and Chemistry                | 40                 |
| MATHSemantics, Physics and Chemistry | 30                 |

- A. The number of students who study only MATHSemantics is 100.
- B. The number of students who study only Physics is 40.
- C. The number of students who study only Chemistry is 40.
- D. The number of students who do not study MATHSemantics, Physics and Chemistry is 70.

Choose the correct answer from the options given below.

- (a) B and D only
- (b) A and B only
- (c) A only
- (d) C only

## MATHS – Set Theory

77. The arithmetic means of two observations is 125 and their geometric means is 60. Find the harmonic mean of the two observations.

- (a) 4.17
- (b) 8.34
- (c) 28.8
- (d) 57.6

## MATHS – A.M.–G.M.–H.M.



78. The arithmetic mean and standard deviation of series of 20 items were calculated by a student as 20 cm and 5 cm respectively. But while calculating them an item 15 was misread as 30. Find the correct standard deviation.

(a) 4.10 (b) 4.40  
(c) 4.54 (d) 4.66

### MATHS – Statistics (Standard Deviation)

79. In a processor, while executing an instruction

- Programme Counter is used to hold the address of next instruction.
- Instruction register holds the instruction for execution.
- Memory Address Register is used to perform address translation.
- Memory Data Register is used to perform data operation.
- Clock generates control signals.

Choose the correct answer form the options given below.

- (a) Only A and B are true  
(b) Only C and D are true  
(c) Only A, B and E are true  
(d) Only B, C and D are true

### COMPUTER – CPU Registers

80. Given the marks of 25 students in the class as  $\{m_1, m_2, \dots, m_{25}\}$ . Marks lie in the range of  $[1 - 100]$  and  $\bar{m}$  is the mean. Which of the following quantity has the value zero?

- (a)  $\sum_{i=1}^{25} |m_i - \bar{m}|$  (b)  $\sum_{i=1}^{25} (m_i - \bar{m})$   
(c)  $\sum_{i=1}^{25} (m_i - \bar{m})^2$  (d)  $\sum_{i=1}^{25} \frac{m_i}{\bar{m}}$

### MATHS – Mean Deviation

81. India has been continuously experiencing military threats from its neighboring countries.  
Statement I: India should engage into an all out war to stop the nagging threats.

Statement II: India should get the neighbors into a serious dialogue to reduce the tension at its border.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both Statement I and Statement II are correct  
(b) Both Statement I and Statement II are incorrect  
(c) Statement I is correct but Statement II is incorrect  
(d) Statement I is incorrect Statement II are correct

### REASONING – Statement Evaluation

82. A + B means A is daughter of B  
A × B means A is son of B  
A – B means A is the wife of B  
P × Q – S means

Statement I: S is father of P.

Statement II: P is daughter of Q.

- (a) Both Statement I and Statement II are true  
(b) Both Statement II and Statement II are false  
(c) Statement I is true but Statement II is false  
(d) Statement I is false but Statement II is true.

### REASONING – Blood Relation

83. The terms  $1, \log_y(x), \log_z(y)$  and  $-15 \log_x(z)$  are in AP. Based on this information answer the following questions.

The common difference of AP is:

- (a) 2 (b) -2  
(c)  $1/2$  (d)  $-1/2$

### MATHS – Logarithmic AP

84. The terms  $1, \log_y(x), \log_z(y)$  and  $-15 \log_x(z)$  are in AP. Based on this information answer the following questions.

The value of  $xy$  is:

- (a) 1 (b) -1 (c)  $z^2$  (d)  $z^3$

### MATHS – Logarithms

85. The terms  $1, \log_y(x), \log_z(y)$  and  $-15 \log_x(z)$  are in AP. Based on this information answer the following questions.

$yz$  is equal to

- (a)  $x$  (b)  $x^2$  (c)  $z^{-2}$  (d)  $z^{-3}$

### MATHS – Logarithms

86. Consider  $n$  events  $E_1, E_2, \dots, E_n$ , with respective probabilities  $p_1, p_2, \dots, p_n$ . If  $(E_1, E_2, \dots, E_n) = \prod_{i=1}^n p_i$  then:

- (a) The events are mutually exclusive  
(b) The events are independent  
(c) The events are dependent  
(d) The events are mutually exclusive and independent

### MATHS – Probability (Independence)

87. Given a set of events  $E_1, \dots, E_n$ , defined on the sample space  $S$  such that:

(i)  $\forall i$  and  $j, i \neq j, E_i \cap E_j = \emptyset$

(ii)  $\bigcup_{i=1}^n E_i = S$

(iii)  $P(E_i) > 0, \forall i = 1, n$

Then the events are:

- (a) Pairwise disjoint and exhaustive  
(b) Pairwise disjoint and independent  
(c) Dependent and mutually exclusive  
(d) Independent and mutually exclusive

### MATHS – Probability (Events)

88. Given the following statement and five possible conclusions.

All Scientists working in America are talented. Some Indian scientists are working in America.

- A. None of Indian scientists are talented.  
B. Some talented Indian scientists have migrated to America.  
C. All talented scientists are Indians.  
D. Some Indian scientists are talented.  
E. Scientists working in India are not talented.

Choose the correct answer from the options given below.

- (a) Conclusions A and D only are correct.  
(b) Conclusions B and D only are correct.  
(c) Conclusions A and E only are correct.  
(d) Conclusions B and E only are correct.

### REASONING – Syllogism



89. Ministers arrived at the public function in their cars. Consider the following statements.

- A. All ministers are rich.
- B. Minister have official cars in general.
- C. Minister usually participate in public functions.
- D. Only rich Ministers arrive in cars.
- E. Person with cars can only be a Minister.

Choose the correct answer from the options given below.

- (a) A and B only
- (b) B and D only
- (c) B and C only
- (d) C and E only

### REASONING – Statement Conclusion

90. The simplified form of Boolean Expression:

$\bar{A}BC + ABC$  is \_\_\_\_\_.

- (a) AC
- (b) AB
- (c) BC
- (d) 0

### COMPUTER – Boolean Algebra

91. 4 Indians, 3 Americans and 2 Britishers are to be arranged around a round table.

Answer the following Questions.

The number of ways of arranging them is

- (a) 9!
- (b)  $9!/2$
- (c) 8!
- (d)  $8!/2$

### MATHS – Circular Permutation

92. 4 Indians, 3 Americans and 2 Britishers are to be arranged around a round table.

Answer the following questions.

The number of ways arranging them so that the two Britishers should never come together is:

- (a)  $7! \times 2!$
- (b)  $6! \times 2!$
- (c) 7!
- (d)  $6! \times {}^7P_2$

### MATHS – Circular Permutation

93. 4 Indians, 3 Americans and 2 Britishers are to be arranged around a round table.

Answer the following questions.

The number of ways of arranging them so that the three Americans should sit together is:

- (a)  $7! \times 3!$
- (b)  $6! \times 3!$
- (c)  $6! {}^6P_3$
- (d)  $6! \times {}^7P_3$

### MATHS – Circular Permutation

94. Match List I with List II

|    | List I                                 |      | List II                 |
|----|----------------------------------------|------|-------------------------|
| A. | The story of my Experiments with Truth | I.   | Maulana Abul Kalam Azad |
| B. | Glimpses of Word History               | II.  | Narndra Modi            |
| C. | India Wins Freedom                     | III. | Pandit Jawaharlal Nehru |
| D. | Exam Warrior                           | IV.  | M. K. Gandhi            |

Choose the correct answer from the options given below:

- (a) A – IV; B – III; C – I; D – II
- (b) A – III; B – IV; C – II; D – I
- (c) A – IV; B – II; C – III; D – I
- (d) A – IV; B – III; C – II; D – I

### GK – Books & Authors

95. Match List I with List II

#### List I

#### List II

- A. Austria I. Oslo
- B. Iran II. Stockholm
- C. Norway III. Vienna
- D. Sweden IV. Tehran

Choose the correct answer from the options given below:

- (a) A – III; B – I; C – IV; D – II
- (b) A – III; B – IV; C – II; D – I
- (c) A – III; B – IV; C – I; D – II
- (d) A – I; B – II; C – III; D – IV

### GK – Capitals

96. Given three identical boxes  $B_1, B_2$  and  $B_3$  each containing two balls.  $B_1$  contains two golden balls,  $B_2$  contains two silver balls and  $B_3$  contains one silver and one golden ball. Conditional probabilities that the golden ball is drawn from  $B_1, B_2, B_3$  are \_\_\_\_\_ respectively.

- (a) 0, 1,  $1/2$
- (b)  $1/2, 0, 1$
- (c) 1, 0,  $1/2$

### MATHS – Conditional Probability

97. Match List I with List II

|    | List I                                                                                                          |      | List II |
|----|-----------------------------------------------------------------------------------------------------------------|------|---------|
| A. | In a GP, the third term is 24 and 6th term is 192. The common ratio is _____.                                   | I.   | 78      |
| B. | Let $S_n$ denote the sum of the first n terms of an AP. If $S_{2n} = 3S_n$ , then $S_{3n}/S_n$ equals to _____. | II.  | 6       |
| C. | The sum of the first 3 terms of a GP is $13/12$ and their product is $-1$ . The first term is _____.            | III. | $-1$    |
| D. | The least value of n for which the sum $3 + 6 + 9 \dots + n$ is greater than 1000 is _____.                     | IV.  | 2       |

Choose the correct answer from the options given below:

- (a) A – III; B – I; C – II; D – IV
- (b) A – III; B – IV; C – I; D – II
- (c) A – IV; B – II; C – III; D – I
- (d) A – IV; B – III; C – II; D – I

### MATHS – AP & GP

98. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

**Assertion A:** Military service should be made compulsory in our country.

**Reason R:** Every citizen should protect his country.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (a) Both A and R are correct and R is the correct explanation of A
- (b) Both A and R are correct and R is not the correct explanation of A
- (c) A is correct but R is not correct.
- (d) A is correct but R is correct.

### REASONING – Assertion-Reason



99. Match List I with List II

|    | List I                                                            |      | List II   |
|----|-------------------------------------------------------------------|------|-----------|
| A. | The value of $\frac{1}{9}(1-w)(1-w^2)(1-w^4)(1-w^8)$ is           | I.   | 0         |
| B. | $w(1+w-w^2)^7$ _____ is equal to                                  | II.  | 1         |
| C. | The least positive integer n such that $(1+w^2)^n = (1+w^4)^n$ is | III. | $-128w^2$ |
| D. | $(1+w-w^2)$ is equal to                                           | IV.  | 3         |

Choose the correct answer from the options given below:

- (a) A – II; B – III; C – I; D – IV  
 (b) A – II; B – III; C – IV; D – I  
 (c) A – III; B – II; C – IV; D – I  
 (d) A – III; B – II; C – I; D – IV

**MATHS – Complex Numbers**

100. Match List I with List II

|    | List I                           |      | List II |
|----|----------------------------------|------|---------|
| A. | $\log_4(\log_3 81) =$            | I.   | 0       |
| B. | $3^{4\log_9 7} = 7^k$ , then K = | II.  | 3       |
| C. | $2^{\log_3 5} - 5^{\log_3 2} =$  | III. | 1       |
| D. | $\log_2[\log_2(256)] =$          | IV.  | 2       |

Choose the correct answer from the options given below:

- (a) A – I; B – III; C – II; D – IV  
 (b) A – I; B – III; C – IV; D – II  
 (c) A – III; B – IV; C – II; D – I  
 (d) A – III; B – IV; C – I; D – II

**MATHS – Logarithms**





# MAARULA CLASSES

TARGET- NIMCET / CUET.PG By: Amit Katiyar (MCA-JNU)

CUET-PG 2022

OFFICIAL PAPER



Scan the QR & Download Our App Now.

DOWNLOAD- ALL THE PAPERS AVAILABLE  
ON THE WEBSITE VISIT AT: [www.Maarula.in](http://www.Maarula.in)

<https://maarula.in/download->

DOWNLOAD OUR APP :  
MAARULA MATHEM -E LEARNING

## ANSWER KEY

| 1.  | 2.  | 3.  | 4.  | 5.  | 6.  | 7.  | 8.  | 9.  | 10.  |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| b   | c   | a   | b   | b   | a   | b   | a   | a   | a    |
| 11. | 12. | 13. | 14. | 15. | 16. | 17. | 18. | 19. | 20.  |
| b   | b   | c   | d   | d   | a   | c   | b   | d   | d    |
| 21. | 22. | 23. | 24. | 25. | 26. | 27. | 28. | 29. | 30.  |
| a   | c   | d   | a   | c   | c   | c   | b   | d   | c    |
| 31. | 32. | 33. | 34. | 35. | 36. | 37. | 38. | 39. | 40.  |
| a   | c   | b   | a   | d   | a   | c   | a   | a   | b    |
| 41. | 42. | 43. | 44. | 45. | 46. | 47. | 48. | 49. | 50.  |
| a   | c   | b   | c   | d   | b   | b   | b   | b   | d    |
| 51. | 52. | 53. | 54. | 55. | 56. | 57. | 58. | 59. | 60.  |
| a   | c   | c   | b   | d   | d   | c   | b   | b   | c    |
| 61. | 62. | 63. | 64. | 65. | 66. | 67. | 68. | 69. | 70.  |
| c   | a   | a   | a   | d   | a   | d   | a   | a   | a    |
| 71. | 72. | 73. | 74. | 75. | 76. | 77. | 78. | 79. | 80.  |
| d   | a   | a   | a   | c   | a   | c   | c   | c   | b    |
| 81. | 82. | 83. | 84. | 85. | 86. | 87. | 88. | 89. | 90.  |
| d   | c   | b   | a   | c   | b   | a   | b   | c   | c    |
| 91. | 92. | 93. | 94. | 95. | 96. | 97. | 98. | 99. | 100. |
| c   | d   | b   | a   | c   | c   | c   | c   | b   | d    |

**ADDRESS:** 117/466, O Block, Geeta Nagar, Sharda Nagar, NEAR: ANURAG HOSPITAL Kanpur, Uttar Pradesh 208025

**website:** [www.question.maarula.in](http://www.question.maarula.in)



**0512-3163515**



**9935985550**



**9554548576**